HỌC VIỆN CÔNG NGHỆ BƯU CHÍNH VIỄN THÔNG

KHOA CÔNG NGHỆ THÔNG TIN 1

BÁO CÁO HÀNG TUẦN

Xây dựng Chatbot hỏi đáp về lịch sử Việt Nam sử dụng GraphRAG

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Weekly Report 8

I. OVERVIEW OF PROGRESS IN THE PAST WEEK

Over the past week, I have completed the entire core backend pipeline for the Chatbot system applying GraphRAG (Retrieval-Augmented Generation combined with Knowledge Graph) technology. Specifically, I have successfully built 3 main modules:

- Knowledge Extraction Module using LLM.
- Graph Construction and Storage Module with Neo4j.
- Query and Q&A Module (GraphRAG Chatbot) applying LangChain.

II. DETAILS OF COMPLETED TASKS

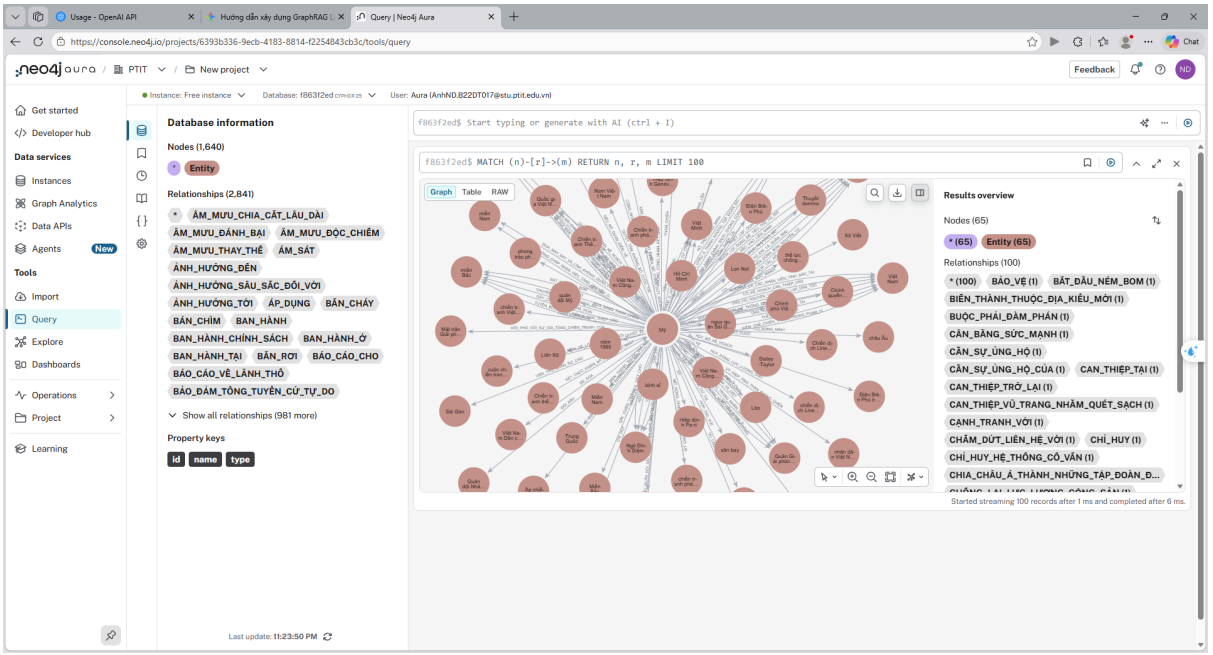
1. Building the Historical Knowledge Extraction Module

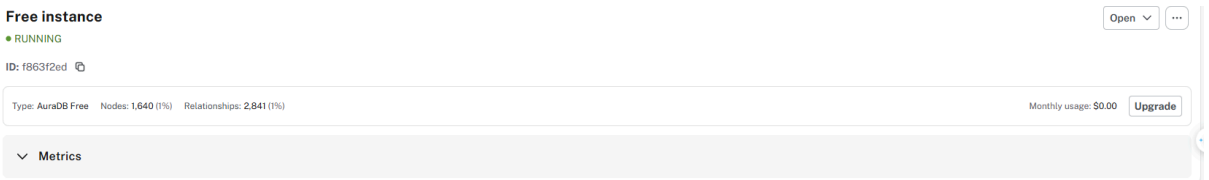
- OpenAI API Integration: Developed a Python script to automatically read historical text segments (processed_chunks.json) and extract Entities (Nodes - Figures, Locations, Organizations, Events, Weapons) along with Relationships (Edges).
- Formatting Assurance (Format Parsing): Applied the response_format={ "type": "json_object" } parameter of the OpenAI gpt-4o-mini model combined with a strict System Prompt, ensuring 100% of the returned results conform to the desired JSON Schema, avoiding data parsing errors.
- Checkpoint & Resume Mechanism: Coded the logic allowing the program to read the incomplete extracted_graph_data.json file. In case of a network error or API interruption, the system automatically saves the processed chunks and resumes from the point of failure upon the next startup, helping to preserve data and optimize API costs.
- AI Parameter Control: Set temperature=0.2 to force the LLM model to focus on extracting objective, accurate information from the original text, minimizing "hallucination" in historical contexts.

```
{} extracted_graph_data.json > {} 0 > {} graph_data > [] nodes > {} 3
36111        {
36194        },
36195        {
36196            "chunk_id": 540,
36197            "source": "wikipedia_khang_chien_chong_my.txt",
36198            "graph_data": {
36199                "nodes": [
36200                    {
36201                        "id": "Quân đội Mỹ",
36202                        "type": "Organization"
36203                    },
36204                    {
36205                        "id": "Quân đội Nhân dân Việt Nam",
36206                        "type": "Organization"
36207                    },
36208                    {
36209                        "id": "Quân Giải phóng miền Nam Việt Nam",
36210                        "type": "Organization"
36211                    },
36212                    {
36213                        "id": "Quân lực Việt Nam Cộng hòa",
36214                        "type": "Organization"
36215                    },
36216                    {
36217                        "id": "Hàn Quốc",
36218                        "type": "Location"
36219                    },
36220                    {
36221                        "id": "Úc",
36222                        "type": "Location"
36223                    },
36224                    {
36225                        "id": "New Zealand",
36226                        "type": "Location"
36227                    },
36228                    {
36229                        "id": "Thái Lan",
36230                        "type": "Location"
36231                    },
36232                    {
36233                        "id": "vũ khí",
36234                        "type": "Event"
```


2. Building the Knowledge Graph Module on Neo4j

- Database Connection: Wrote a script to securely connect to the Neo4j database using the neo4j library and environment variables (.env).
- Data Cleaning Mechanism: Completely resolved the Cypher syntax error issue when the relationship type contains special characters. The current code automatically converts to uppercase, removes commas, and replaces spaces and hyphens with underscores (_).
- Node/Edge Initialization Algorithm: Used the MERGE command instead of CREATE in Cypher to insert Entities and Relationships. This ensures idempotency, preventing the creation of duplicate nodes (e.g., the figure "Ho Chi Minh" appearing in multiple chunks will be consolidated into a single Node on the graph).
- The database has 1640 nodes and 2841 relationships.





3. Developing the GraphRAG Chatbot Module Integrated with LangChain

- LangChain & Neo4jGraph Integration: Replaced manual queries with LangChain's GraphCypherQAChain, directly connecting the gpt-4o-mini "brain" with the Neo4j database.
- Prompt Template Optimization (Text-to-Cypher): This is a significant step forward. The team designed a specialized CYPHER_GENERATION_TEMPLATE for Vietnamese historical data:
    - Fuzzy Search: Solved the issue of users inputting queries with inaccurate capitalization using the toLower() and CONTAINS functions.
    - Semantic Relationship Querying: Guided the AI on how to map natural language questions (e.g., "What is the plot of the US?") to searching for texts containing equivalent keywords in the relationship names.
- Basic Interactive Interface: Built a communication loop (Terminal CLI) allowing students to chat directly with the system, testing the quality of answers in real-time.

```
✅ KẾT NỐI THÀNH CÔNG!
==========================================================
🤖 PHẦN MỀM HỌC LỊCH SỬ VIỆT NAM - CHATBOT ĐÃ SẴN SÀNG
==========================================================

🧑 Sinh viên: âm mưu của Mỹ tại Việt Nam là gì
🤖 Chatbot đang tra cứu đồ thị tri thức...


> Entering new GraphCypherQAChain chain...
Generated Cypher:
cypher
MATCH (n:Entity)-[r]->(m:Entity)
WHERE toLower(n.id) CONTAINS 'mỹ' AND type(r) CONTAINS 'ÂM_MƯU'
RETURN n.id AS `Chủ thể`, type(r) AS `Hành động`, m.id AS `Đối tượng`

Full Context:
[{'Chủ thể': 'Mỹ', 'Hành động': 'ÂM_MƯU_THAY_THẾ', 'Đối tượng': 'Pháp'}, {'Chủ thể': 'Mỹ', 'Hành động': 'ÂM_MƯU_ĐỘC_CHIẾM', 'Đối tượng': 'Đông Dương'}, {'Chủ thể': 'đế quốc Mỹ', 'Hành động': 'ÂM_MƯU_THAY_THẾ', 'Đối
ợng': 'Pháp'}, {'Chủ thể': 'đế quốc Mỹ', 'Hành động': 'ÂM_MƯU_ĐỘC_CHIẾM', 'Đối tượng': 'Đông Dương'}, {'Chủ thể': 'Đế quốc Mỹ', 'Hành động': 'ÂM_MƯU_ĐÁNH_BẠI', 'Đối tượng': 'Cách mạng miền Nam'}]

> Finished chain.

🤖 Chatbot: Âm mưu của Mỹ tại Việt Nam bao gồm việc âm mưu thay thế Pháp, âm mưu độc chiếm Đông Dương, và âm mưu đánh bại cách mạng miền Nam.
```

III. PLANS AND FUTURE DEVELOPMENT DIRECTIONS

- Accuracy Evaluation: Prepare a test set of historical questions (QA Dataset) to evaluate the accuracy of the Cypher queries generated by the AI.
- Vector Database Integration (Optional): Combine Vector/Embeddings search for long-form content (detailed event descriptions) in parallel with Graph search to provide richer system responses.
- Frontend Development: Transition the interface from a Terminal CLI to a complete Web App (using Streamlit, Flask, or a combination of React/Vue depending on the team's direction) for integration into the final software.